

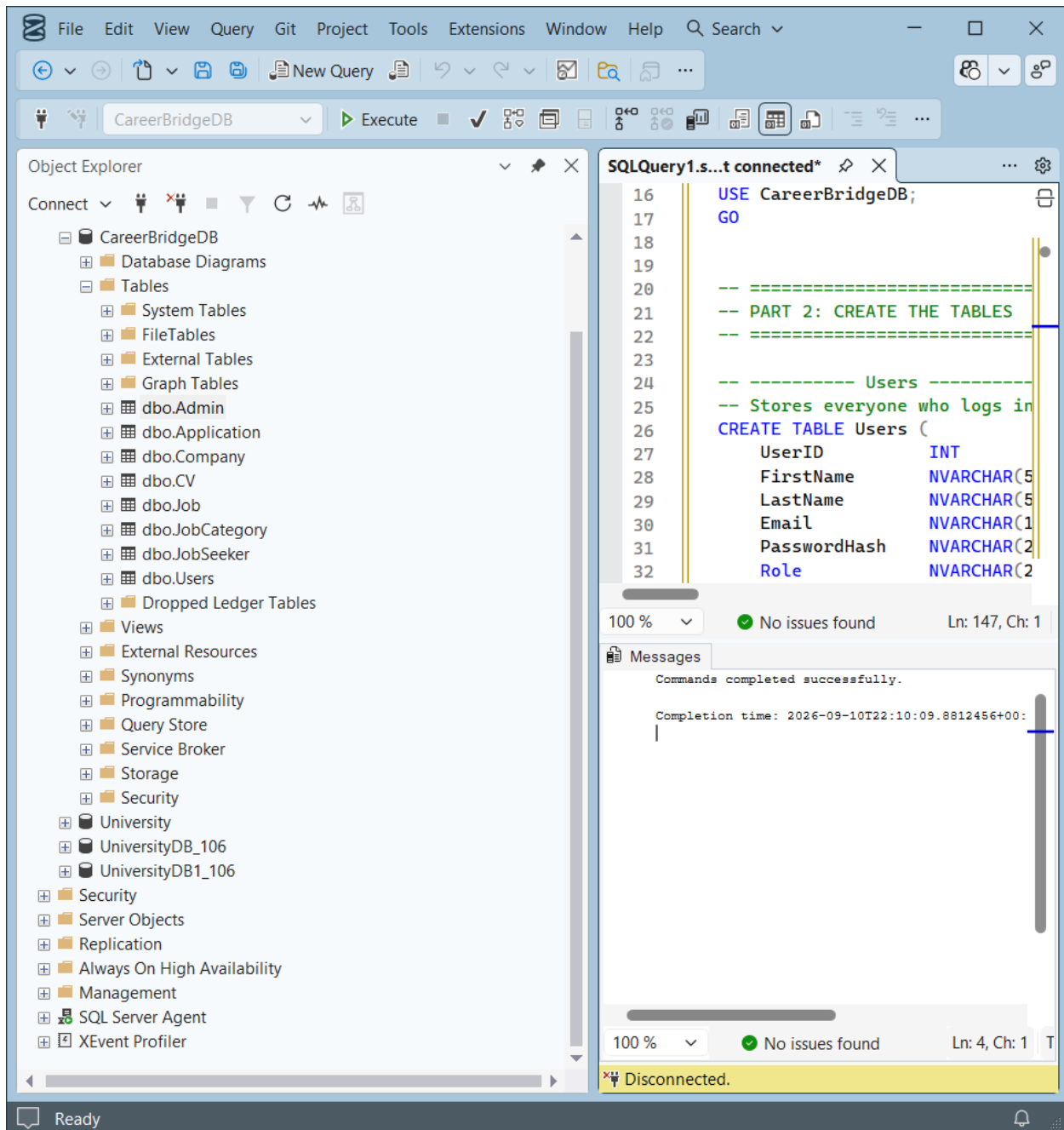
CAREERBRIDGE DAY 4 EXERCISE

Database Creation and Core SQL Practice

Name: Larbi Peniels Okyerewa

Date: 10th September 2026

SCREENSHOT 1: Database and Tables in SSMS



SCREENSHOT 2: Successful UPDATE

File Edit View Query Git Project Tools Extensions Window Help Search

CareerBridgeDB Execute

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SQLQuery1....95\G6 (53))*

DESKTOP-M20J... - dbo.Users

Results Messages

UserID	FirstName	LastName	Phone
1	1	Larbi	Okyerewa

JobID	JobTitle	IsActive
1	6	Accounting Assistant

ApplicationID	JobSeekerID	JobID	Status
1	1	1	Shortlisted

CompanyID	CompanyName	CompanyWebsite
1	1	TechGhana Solutions

DESKTOP-M20JB95 (17.0 RTM) | DESKTOP-M20JB95\G6 (53) | CareerBridgeDB | 00:00:00 | Row: 1, Col: 1 | 4 rows

Ready

SCREENSHOT 3: JOIN Query Result

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CareerBridgeDB Execute

SQLQuery1....95\G6 (53)* DESKTOP-M20J... - dbo.Users

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Results Messages

	JobTitle	CompanyName	Location	JobType	ApplicationDeadline
1	Software Developer Intern	TechGhana Solutions	Accra	Internship	2026-03-15 00:00:00.000
2	Data Analyst Intern	DataWorks Africa	Accra	Internship	2026-03-20 00:00:00.000
3	Network Engineer	NetLink Systems	Kumasi	Full-Time	2026-04-10 00:00:00.000
4	Junior Software Developer	TechGhana Solutions	Accra	Full-Time	2026-04-15 00:00:00.000
5	Business Intelligence Analyst	DataWorks Africa	Accra	Part-Time	2026-03-30 00:00:00.000
6	Accounting Assistant	NetLink Systems	Accra	Graduate Programme	2026-04-25 00:00:00.000

	ApplicationID	ApplicantName	JobID	ApplicationDate	Status
1	1	Larbi Okyerewa	1	2026-01-09 00:00:00.000	Shortlisted
2	2	Abena Danso	2	2026-01-11 00:00:00.000	Shortlisted
3	3	Kwame Appiah	3	2026-01-13 00:00:00.000	Submitted
4	4	Akosua Boate...	6	2026-01-21 00:00:00.000	Submitted
5	5	Nana Adjei	5	2026-01-19 00:00:00.000	Shortlisted
6	6	Larbi Okyerewa	4	2026-01-16 00:00:00.000	Submitted
7	7	Abena Danso	5	2026-01-20 00:00:00.000	Submitted
8	8	Kwame Appiah	1	2026-01-22 00:00:00.000	Rejected

	ApplicantName	JobTitle	CompanyName	ApplicationDate	Status
1	Larbi Okyerewa	Software Developer Intern	TechGhana Solutions	2026-01-09 00:00:00.000	Shortlisted
2	Abena Danso	Data Analyst Intern	DataWorks Africa	2026-01-11 00:00:00.000	Shortlisted
3	Kwame Appiah	Network Engineer	NetLink Systems	2026-01-13 00:00:00.000	Submitted
4	Akosua Boate...	Accounting Assistant	NetLink Systems	2026-01-21 00:00:00.000	Submitted
5	Nana Adjei	Business Intelligence An...	DataWorks Africa	2026-01-19 00:00:00.000	Shortlisted
6	Larbi Okyerewa	Junior Software Develop...	TechGhana Solutions	2026-01-16 00:00:00.000	Submitted
7	Abena Danso	Business Intelligence An...	DataWorks Africa	2026-01-20 00:00:00.000	Submitted
8	Kwame Appiah	Software Developer Intern	TechGhana Solutions	2026-01-22 00:00:00.000	Rejected

	JobTitle	CategoryName
1	Software Developer Intern	Software Development
2	Data Analyst Intern	Data & Analytics
3	Network Engineer	Networking
4	Junior Software Develop...	Software Development
5	Business Intelligence An...	Data & Analytics
6	Accounting Assistant	Accounting

DESKTOP-M20JB95 (17.0 RTM) | DESKTOP-M20JB95\G6 (53) | CareerBridgeDB | 00:00:00 | Row: 1, Col: 1 | 28 rows

Ready

SCREENSHOT 4: GROUP BY Query Result

File Edit View Query Git Project Tools Extensions Window Help Search

CareerBridgeDB Execute

SQLQuery1....95\G6 (53)* DESKTOP-M20J... - dbo.Users

Results Messages

TotalUsers	
1	5

TotalJobs	
1	6

ActiveJobs	
1	5

	JobTitle	TotalApplications
1	Business Intelligence Analyst	2
2	Software Developer Intern	2
3	Accounting Assistant	1
4	Data Analyst Intern	1
5	Junior Software Developer	1
6	Network Engineer	1

	CompanyName	TotalJobs
1	DataWorks Africa	2
2	NetLink Systems	2
3	TechGhana Sol...	2

	Status	Total
1	Submitted	4
2	Shortlisted	3
3	Rejected	1

DESKTOP-M20JB95 (17.0 RTM) | DESKTOP-M20JB95\G6 (53) | CareerBridgeDB | 00:00:00 | Row: 1, Col: 1 | 15 rows

Ready

SCREENSHOT 5: Challenge Queries

SQLQuery1....95\G6 (53)*

DESKTOP-M20J... - dbo.Users

Results Messages

	CategoryID	JobTitle	Description	Location	JobType
1	4	Accounting Assistant	Graduate programme for accounting graduates inte...	Accra	Graduate Progra
2	2	Business Intelligence Analyst	Part-time role creating Power BI dashboards and re...	Accra	Part-Time
3	1	Junior Software Developer	Full-time role building and maintaining web applicat...	Accra	Full-Time
4	3	Network Engineer	Design, install and maintain network infrastructure f...	Kumasi	Full-Time
5	2	Data Analyst Intern	Assist our analytics team with data cleaning, reportin...	Accra	Internship

	ApplicationID	ApplicantName	JobTitle	ApplicationDate	Status
1	1	Larbi Okyerewa	Software Developer Intern	2026-01-09 00:00:00.000	Shortlisted
2	6	Larbi Okyerewa	Junior Software Developer	2026-01-16 00:00:00.000	Submitted
3	8	Kwame Appiah	Software Developer Intern	2026-01-22 00:00:00.000	Rejected

	ApplicationID	JobTitle	CompanyName	ApplicationDate	Status
1	1	Software Developer Intern	TechGhana Solutions	2026-01-09 00:00:00.000	Shortlisted
2	6	Junior Software Developer	TechGhana Solutions	2026-01-16 00:00:00.000	Submitted

JobTitle	TotalApplications

CompanyName	TotalJobs
1 NetLink Systems	2

DESKTOP-M20JB95 (17.0 RTM) | DESKTOP-M20JB95\G6 (53) | CareerBridgeDB | 00:00:00 | Row: 1, Col: 1 | 11 rows

PART 4 — PRIMARY KEYS AND FOREIGN KEYS

For each table, identify:

- Primary Key
- Foreign Key(s)

Complete the table below.

Table	Primary Key	Foreign Key(s)
User	UserID	None
JobSeeker	JobSeekerID	UserID
Admin	AdminID	UserID
Company	CompanyID	None
Job Application	JobID	CompanyID,
CategoryID	ApplicationID	JobSeekerID, JobID
Job Category	CategoryID	None
CV	CVID	JobSeekerID

PART 14 — DATABASE RULES

Your database should enforce important rules where possible.

Think about the following:

Email Address

Should two users be able to register with the same email address?

Answer:

No

What database rule could help prevent this?

A UNIQUE Constraint should be placed on the email column of the users.

Duplicate Applications

Should one Job Seeker be able to apply for the same job twice?

Answer:

NO

How could the database help prevent this?

A CONSTRAINT can be placed across multiple fields together. If a candidate tries to submit another row containing an identical combination, SQL Server blocks the entry.

Foreign Keys

Should an application be allowed to reference a JobID that does not exist?

Answer:

No

Explain why:

Because of the foreign key, an application cannot link to a non-existent JobID.

PART 15 — THINKING QUESTIONS

Answer briefly in your own words.

1. What is the difference between DELETE and DROP?

Delete only removes the rows inside a table but drops delete the whole table.

2. What is the difference between WHERE and HAVING?

WHERE acts as a filter on individual rows before rows are grouped together whiles HAVING acts as a filter on grouped summary values after a GROUP BY clause runs.

3. Why do we use JOINS?

To link two tables together.

4. What is the difference between INNER JOIN and LEFT JOIN?

INNER JOIN only outputs rows where records exist on both sides but LEFT JOIN displays every row from the left table regardless of whether the right table contains matching entities.

5. Why are Foreign Keys important?

They secure relationships between independent tables, preventing broken relationships, tracking child logs, and blocking invalid rows from entering tables.

6. Why is it better to store CompanyID in the Job table instead of repeatedly storing CompanyName?

Data normalization. Storing CompanyID avoids text duplication. If a company updates its name, it can be corrected in **one row** within the Companies table instead of scanning and changing thousands of separate job posts.

7. What is the purpose of GROUP BY?

It collapses multiple duplicate data rows into summary blocks based on matching fields so that aggregate math operations (COUNT, SUM, AVG) can be executed across categories.

8. What could happen if the wrong data type is selected for a column?

It degrades processing speeds, exhausts storage drive bytes, and leads to application bugs.